Towards Multi-Permeability Cores Through Composition-Controlled Pellet Extrusion

1 Background

Reducing the volume of inductors and transformers is an important route to increasing power density in power converters [1]. Their soft magnetic cores guide magnetic flux and influence electromagnetic performance [2]. The magnetic flux distribution within a core depends on the geometry, material properties and operating conditions. Nonuniform flux density can lead to uneven core losses and localized heating [3]. In some inductor designs, regions near the conductor approach magnetic saturation while material farther away remains underutilized [4]. Near saturation, further increases in magnetising field produce relatively little additional flux density as the material's incremental permeability decreases. This can reduce the effective inductance as the current increases, limiting the utilisation of the core [2, 5]. These limitations motivate designs that distribute magnetic flux more effectively.

One approach is to optimize the core geometry and winding arrangement. For example, Cui et al. designed an inductor with an improved flux distribution by shaping the core and distributing the winding turns [6]. A complementary approach is to vary permeability spatially within the core to tailor the flux distribution [4, 7]. Wang et al. experimentally demonstrated that a three-permeability inductor achieved higher inductance than a single-permeability counterpart and improved the efficiency of the tested high-frequency DC/DC converter [4]. Liu et al. reported that a three-permeability toroidal core achieved the same inductance at a smaller volume than a single-permeability core [7]. These results demonstrate the potential benefits of spatial permeability variation for specific designs, although the benefits depend on the core geometry and operating conditions. Geometry and permeability are not mutually exclusive design variables, and optimising both may offer further benefits [8].

Several manufacturing approaches have been investigated to realize spatially varying permeability. Lu et al. produced discrete radial permeability regions by compacting individual rings at different pressures [9]. Continuous radial permeability variation has also been demonstrated by imposing a temperature gradient during sintering [10]. These approaches demonstrate the feasibility of creating multi-permeability cores, but the reported implementations rely on specific core geometries and manufacturing arrangements.

Additive manufacturing offers an alternative for producing customized geometries and positioning different material formulations within a component. Several AM processes have been investigated for soft-magnetic materials, for example powder-bed fusion, binder jetting and material extrusion [11, 12]. The suitability of each process depends on the feedstock, required magnetic properties and post-processing conditions. Material extrusion is particularly relevant for thermoplastic polymers and composites because it offers relatively low initial investment, simple operation and the capacity for simultaneous use of multiple materials [13].

Liu et al. demonstrated that material extrusion can be used to manufacture multi-permeability cores. They employed a multi-extruder paste-based 3D printer to fabricate a toroidal core with three distinct permeability regions [7]. Their ferrite paste required sintering above 900 °C, whereas their powdered-iron paste could be cured below 250 °C. This demonstrates the feasibility of additive manufacturing for multi-permeability cores, while illustrating the importance of feedstock formulation and post-processing requirements.

Within material extrusion, the choice of feedstock involves a trade-off between magnetic performance and manufacturability. Producing filament-based composites with high magnetic-particle loading while retaining suitable processability is challenging [14]. Cañada et al. report that filament-based soft-magnetic cores in the studies they reviewed generally exhibited relative permeabilities below 10. Paste-based approaches can achieve much higher permeabilities, but their rheology can complicate dimensional accuracy, and some formulations require

high-temperature sintering. In contrast, pellet-based feedstock can provide moderate permeability without high-temperature post-processing. Cañada et al. demonstrated nylon-12/FeSiAl composite cores with a relative permeability of approximately 42 using pellet-based material extrusion [15]. Since these results come from different materials and studies, they do not constitute a controlled comparison of the three feedstock types. Nevertheless, pellet extrusion provides a promising starting point for this work because it has already demonstrated useful soft-magnetic properties without high-temperature post-processing. Additionally, Cañada et al. reported variation in permeability for printed nylon-12/FeSiAl composites between different frequencies. Therefore, the permeability of the printed composites should be characterised at defined frequencies to establish whether changes in pellet blend ratio affect permeability.

The composition of a polymer-based magnetic composite influences its magnetic properties. Hu et al. experimentally investigated the permeability of polymer composites with different magnetic inclusion contents, although their matrix differed from the nylon-12 composite considered here [16]. Huber et al. demonstrated variable-composition printing by blending NdFeB-filled PA12 filament with unfilled PA12 filament. Their approach controlled the remanent magnetization distribution of hard-magnetic permanent magnets to produce a predefined stray field, rather than the effective permeability of a soft-magnetic core [17]. This demonstrates that magnetic composition can be varied during material extrusion, but does not establish how pellet blend ratio affects the permeability of a soft-magnetic composite.

Composition-controlled pellet extrusion has also been investigated in non-magnetic material systems. Brackett et al. studied in-situ blending of neat and carbon-fibre-filled ABS pellets and showed that transitions between compositions occupy a finite distance and depend on the direction of the material change [18]. This indicates that changes in the input blend ratio may not translate immediately into corresponding changes in the extruded material. However, the material system and large-format extrusion process they investigated differ from the present application. Elsersawy et al. found that dedicated mixing elements improved the uniformity of blended and colored PLA compared with a standard Mahor V4 pellet extruder without a mixing element [19]. Although this demonstrates the importance of mixing configuration, colour uniformity does not establish uniform magnetic filler distribution or reproducible permeability. It therefore remains unclear whether a standard pellet extruder provides sufficient mixing to produce predictable and repeatable magnetic properties when blending magnetic-filled and unfilled polymer pellets.

2 Problem Statement

The practical challenge is to determine whether blending magnetic-filled and unfilled polymer pellets using a standard pellet extruder provides a reliable means of controlling the effective permeability of the printed material. The nominal blend ratio may not correspond directly to the local composition of the extruded material, while residual material can complicate transitions between different blends during printing. This work will therefore characterize the blend-ratio–permeability relationship, assess mixing-related variation, and evaluate whether a controlled pause–purge–resume procedure can be used to manufacture a core with three distinct permeability regions.

2.1 Research Question

To address the discovered problems, the work will investigate the following research questions:

> *Can varying the blend ratio of magnetic-filled and unfilled polymer pellets produce reproducible changes in the permeability of a soft-magnetic composite manufactured using a commercial pellet extruder?*

Can discrete permeability regions be fabricated by sequentially printing pre-blended pellet batches, using controlled purging to achieve distinct and reproducible material compositions?

2.2 Requirements

To address the research question and capture the overarching scientific plan, the following requirements have been defined:

- Establish a repeatable pellet-extrusion process for producing test specimens from the selected magnetic-filled and unfilled polymer pellets.
- Manufacture three specimen per each magnetic-filled-to-unfilled pellet mass ratio (0:100, 25:75, 50:50, 75:25 and 100:0).
- Characterize the effective permeability and its variability between each specimen in a given ratio.
- Characterize the amount of magnetic filler present in each of the specimen.
- Characterize the uniformity of magnetic filler in each of the specimen.
- Assess the correlation between effective permeability, amount of magnetic filler present and the uniformity of magnetic filler.
- Establish a repeatable pause-purge-resume procedure that limits residual material from the previous pellet batch when switching blend ratios.
- Evaluate the feasibility of fabricating a multi-permeability core with three distinct permeability regions by sequentially printing pre-blended pellet batches.
- Characterize the multi-permeability core's inductance and, where resources permit, saturation behaviour and core losses, and compare its performance with a uniform-permeability reference core of identical geometry.

2.3 Methodology

To fulfill the set-out requirements, the following methodology has to be used:

- Conduct a targeted review of commercially available pellet extruders and compatible polymer feedstocks. Create an evaluation matrix, setting out project criterion with their specific weight. Select the highest rated path of solutions. Develop and assemble the selected manufacturing process. Investigate appropriate printing parameters.
- Design a specimen geometry. Using the developed manufacturing process, fabricate three specimen at each specific (0:100, 25:75, 50:50, 75:25, 100:0) magnetic-filled-to-unfilled pellet mass ratios.
- Wind the same number of turns of isolated copper wire around each specimen, ensuring consistent winding arrangement. Measure the inductance and its between-specimen variability using an impedance analyzer across a specific frequency range. Calculate the effective permeability.
- Use Thermogravimetric Analysis (TGA) under inert gas (to avoid oxidation) to estimate bulk filler content, calibrated to the known filler content. Samples from independent extrusion runs and different specimen locations will be analyzed.

- Use Scanning Electron Microscopy paired with Energy-Dispersive X-ray Spectroscopy (SEM–EDS) mapping of cross-sections to quantify local FeSiAl distribution. Samples from independent extrusion runs and different specimen locations will be analyzed. Use image processing to quantify numerically the filler distribution.
- Plot the results from TGA, SEM-EDS and effective permeability. Visually inspect whether there are trends. Draw a conclusion.
- To determine the required purge amount, a print will be paused after reaching steady-state extrusion with a 0:100 blend, and the remaining pellets will be removed from the hopper before loading the 100:0 blend. The extruder will then be purged into separately collected samples at defined cumulative extruded masses. TGA will be used to estimate the composition of each sample, using known blend ratios to establish a calibration. The composition will be plotted against the cumulative purge mass to determine how much material must be purged before the new blend reaches a stable composition. The procedure will be repeated to additionally for the transition from 100:0 to 0:100 ratio.
- Design a multi-permeability magnetic core with three regions (100:0, 75:25 and 50:50) arranged to accommodate sequential material changes. Assign each region a blend ratio based on the experimentally characterized permeability values. Manufacture the core using the established pause-purge-resume procedure. Visually assess whether the intended regions have distinct compositions.
- Wind the same amount of turns of isolated copper wire around the manufactured core. Measure the inductance of the multi-permeability specimen and uniform permeability specimen with an impedance analyzer across a specific frequency range. Remove the winding. Wind an excitation coil and a separate sensing coil around the core. Drive the excitation coil with a controlled AC source. Measure the excitation current and sensing coil voltage. Calculate the magnetic field strength and flux density. To measure the core losses, adjust the source, so the core reaches the peak flux density. Record the B-H loop. Calculate the energy loss from its enclosed area. Repeat the procedures for uniform permeability specimen. Compare the results.

2.4 Delimitations

The project shall use the available commercial pellet extruder and the selected magnetic-filled and unfilled polymer feedstocks as the baseline. The project will not investigate the effect density and porosity has on the effective permeability. The development of an automated, continuously adjustable multi-material feeding system is outside the initial scope.

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